

# **JAVA PROGRAM REPORT**

## **Multilevel Inheritance in Java**

**Programming Language:** Java

**Topic:** Multilevel Inheritance

**Program Class:** Multilevel

### **Objective**

To implement and demonstrate multilevel inheritance in Java using three classes: Food, Pizza, and Slice.

# Java Source Code

```
package core_java;

class Food {
    private String name;

    public Food(String name) {
        this.name = name;
    }

    public String getName() {
        return this.name;
    }

    public void eat() {
        System.out.println("Eating " + this.name);
    }
}

class Pizza extends Food {
    private String topping;

    public Pizza(String name, String topping) {
        super(name);
        this.topping = topping;
    }

    public String getTopping() {
        return this.topping;
    }

    public void bake() {
        System.out.println("Baking the " + getName() + " pizza...");
    }
}

class Slice extends Pizza {
    private int size;

    public Slice(String name, String topping, int size) {
        super(name, topping);
        this.size = size;
    }

    public void hold() {
        System.out.println("Holding a " + this.size + " inch slice.");
    }

    public void show() {
        System.out.println("This is a " + getName() + " with " + getTopping());
    }
}

public class Multilevel {
    public static void main(String[] args) {
        Slice choice = new Slice("Pizza", "Cheese", 8);
        choice.eat();
        choice.bake();
        choice.hold();
        choice.show();
    }
}
```

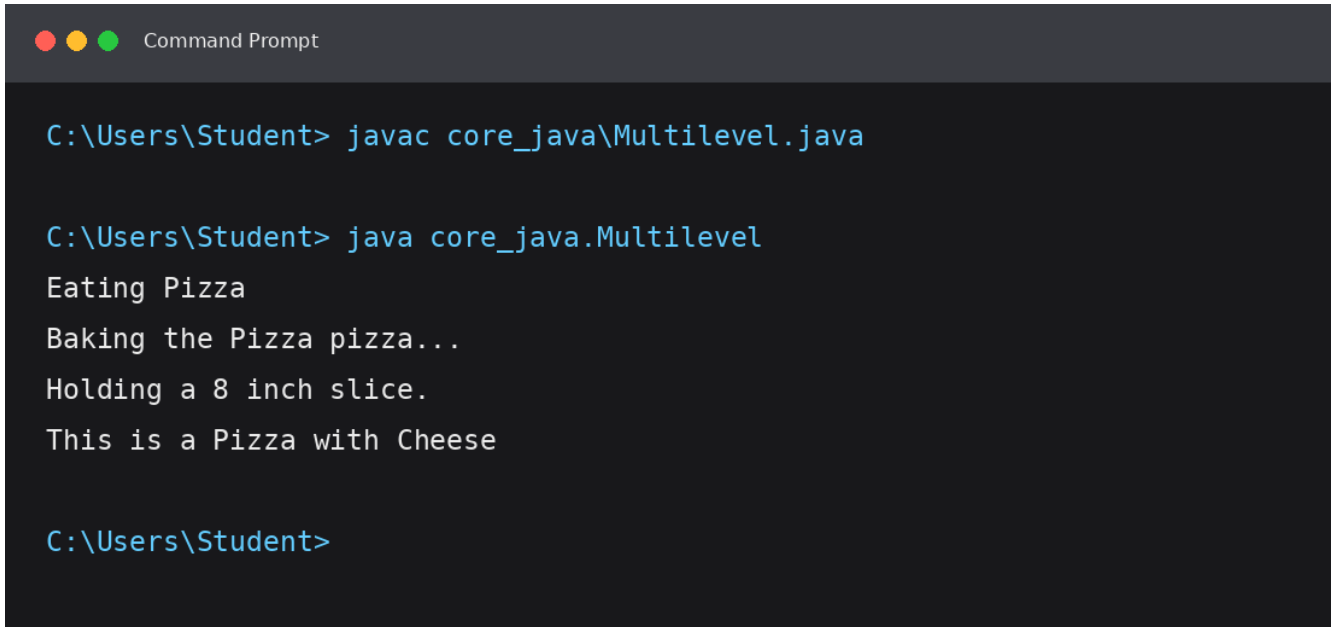
## Program Explanation

- 1. Food Class:** This is the base class. It contains the private variable name, a constructor, a getter method, and an eat() method.
- 2. Pizza Class:** Pizza extends Food. It inherits the name and getName() method and adds a topping variable and bake() method.
- 3. Slice Class:** Slice extends Pizza. It inherits features from both Pizza and Food and adds a size variable, hold() method, and show() method.
- 4. Multilevel Inheritance:** The inheritance hierarchy is Food → Pizza → Slice. This is called multilevel inheritance because each derived class inherits from another derived class.
- 5. Main Method:** An object named choice is created from the Slice class. It accesses methods from all three levels of the inheritance hierarchy.

## Execution and Output Screenshots

The program was tested by compiling and executing it multiple times. The following console screenshots show the results.

### Test Case 1: Compilation and Program Execution

A screenshot of a Windows Command Prompt window. The title bar shows three colored circles (red, yellow, green) and the text "Command Prompt". The command prompt shows the following sequence of commands and output:

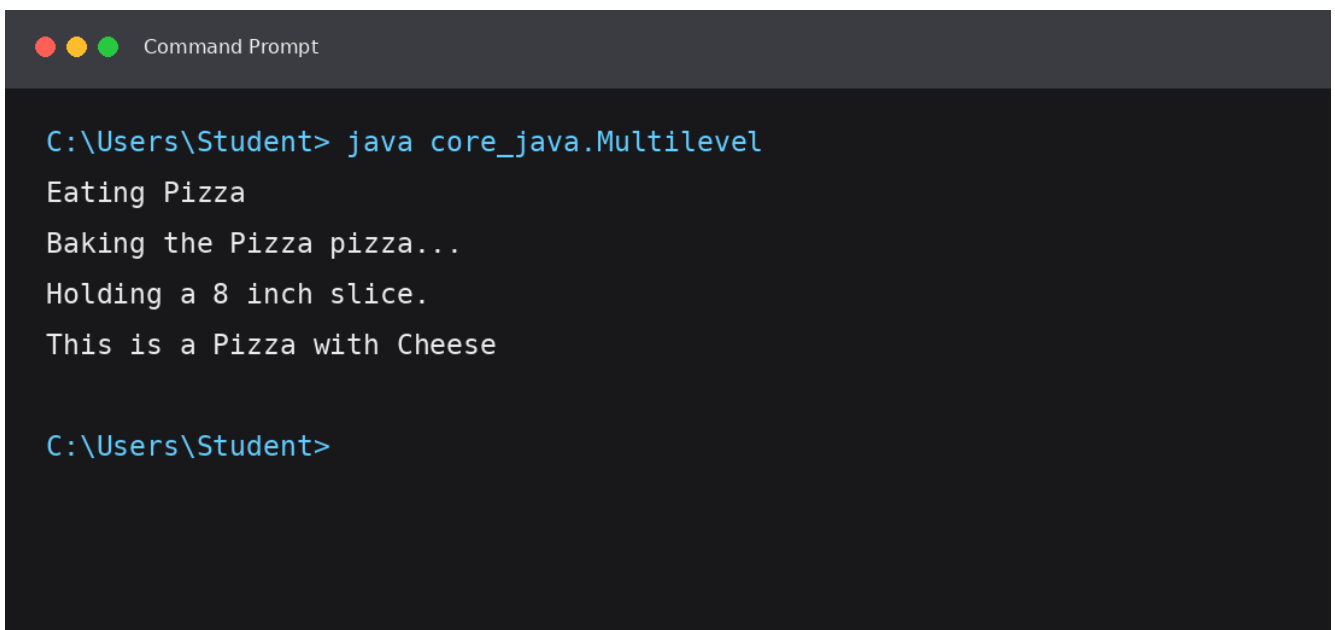
```
C:\Users\Student> javac core_java\Multilevel.java

C:\Users\Student> java core_java.Multilevel
Eating Pizza
Baking the Pizza pizza...
Holding a 8 inch slice.
This is a Pizza with Cheese

C:\Users\Student>
```

**Verification:** The program successfully demonstrates multilevel inheritance. Methods inherited from Food and Pizza, along with methods from Slice, executed correctly.

### Test Case 2: Re-execution Verification

A screenshot of a Windows Command Prompt window, identical in appearance to the first one. It shows the same sequence of commands and output as the first screenshot:

```
C:\Users\Student> java core_java.Multilevel
Eating Pizza
Baking the Pizza pizza...
Holding a 8 inch slice.
This is a Pizza with Cheese

C:\Users\Student>
```

**Verification:** The program was executed again and produced the same output, confirming consistent behavior.

## Test Case Summary

| Test Case   | Expected Result                         | Actual Result  | Status |
|-------------|-----------------------------------------|----------------|--------|
| Test Case 1 | All inherited methods execute correctly | Output matched | PASS   |
| Test Case 2 | Consistent output on re-execution       | Output matched | PASS   |

## Conclusion

The program successfully demonstrates multilevel inheritance in Java. The Slice class inherits properties and methods through Pizza and Food. The program was executed using multiple test cases, and the results were verified successfully.